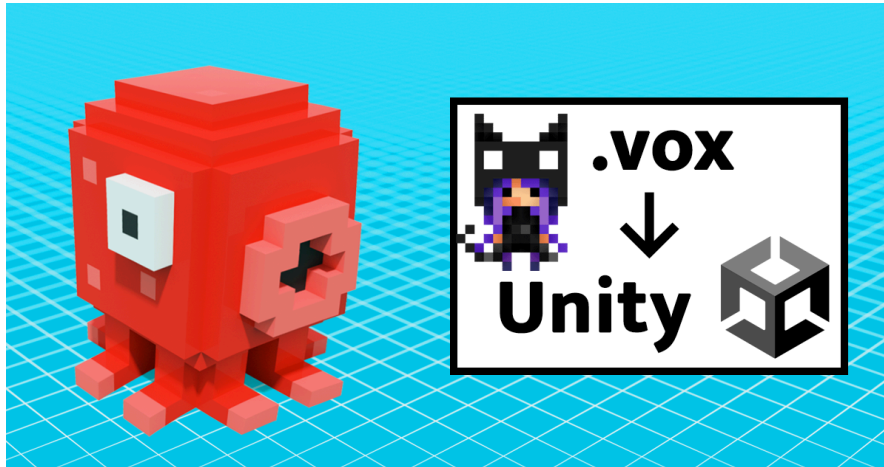


Voxel Toolkit: MagicaVoxel Importer & Optimizer



Efficiently import your MagicaVoxel creations into Unity with optimized mesh and texture generation and customizable settings. Simply drop .vox files into your project for an intuitive workflow.

If you have any questions about Voxel Toolkit, please email pirate.parrot.software@gmail.com or join our [Discord channel](#). When reporting a bug, please include steps to reproduce it. Keep in mind that we receive many messages daily, so please be patient — we'll respond as soon as possible. Additionally, if you have suggestions for new features, let us know — many of our best features came from your ideas!

- [API documentation](#)
- [Overview of Voxel Toolkit](#)
- [Quick Demo](#)
- [Runtime Modification Demo](#)
- [Game Demo](#)
- [Example of the game made with the asset](#)

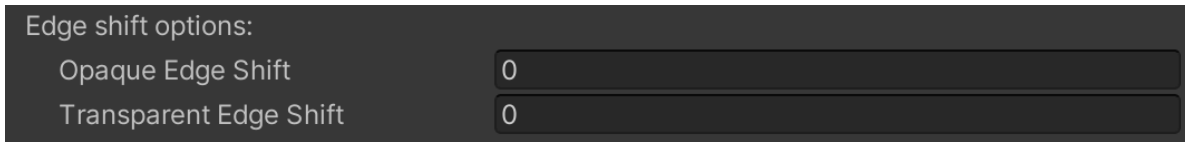
★ We'd Love to Hear Your Feedback! ★

If you enjoy using Voxel Toolkit: MagicaVoxel Importer & Optimizer, please consider leaving a review on the [Unity Asset Store](#)! Your ratings and feedback help us improve the asset and continue to develop new features.

Thank you for your support! 🙏

Parameters for Importing .vox Files

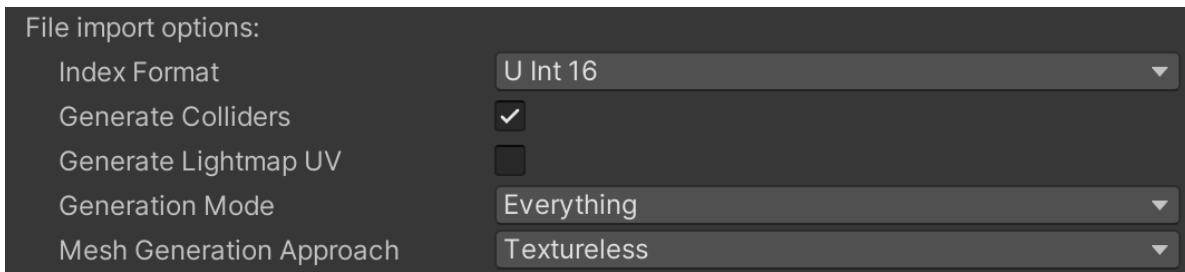
⚠ Please note that in order for the asset to store objects, it's better to name items inside .vox files with unique names as MagicaVoxel doesn't provide any stable ID for us to use. ⚠



Edge shift options:

Opaque Edge Shift	0
Transparent Edge Shift	0

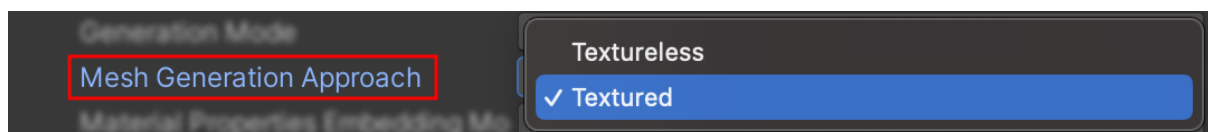
1. **Opaque Edge Shift** and **Transparent Edge Shift** determine how far the quad boundaries are offset during generation.



File import options:

Index Format	U Int 16
Generate Colliders	☒
Generate Lightmap UV	☐
Generation Mode	Everything
Mesh Generation Approach	Textureless

2. **Index Format** determines the type of indexing used for meshes. Select UInt32 for 32-bit unsigned indexing, or UInt16 for 16-bit unsigned indexing, depending on your project's requirements and target platform.
3. **Generate Colliders** determines whether the asset should automatically create mesh colliders for the generated objects.
4. **Generate Lightmap UV** determines whether the asset should automatically generate UV coordinates for lightmapping on the meshes.
5. **Generation Mode** specifies the type of assets to be generated. If 'EssentialOnly' is selected, only Unity assets (models, transforms, etc.) are generated without actual voxel data.
6. **Mesh Generation Approach** determines whether the mesh will use a generated texture or embed material data directly into its vertices. If the **Textured** mode is selected, two additional options (described below) will become available.

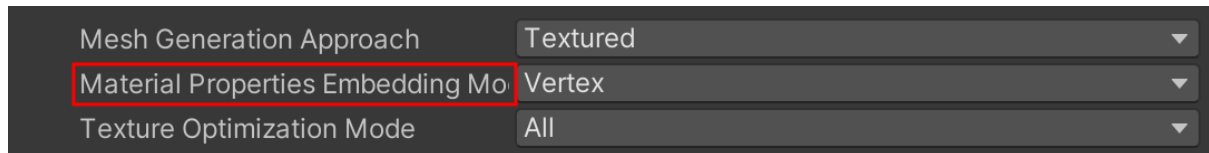


Generation Mode

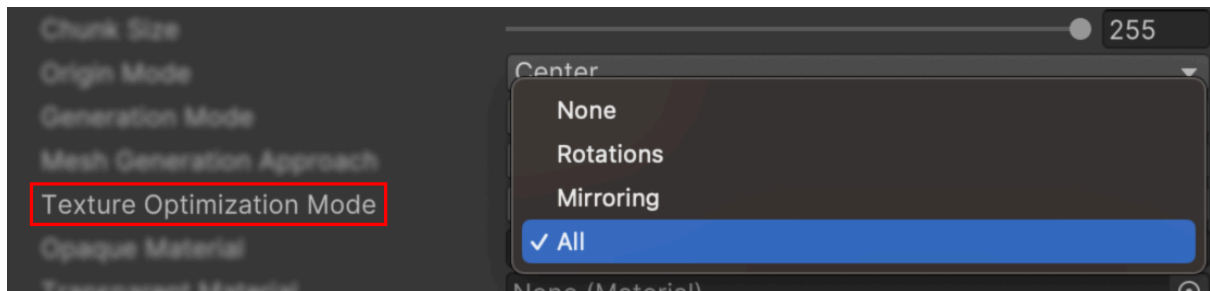
Mesh Generation Approach

Textureless

✓ Textured

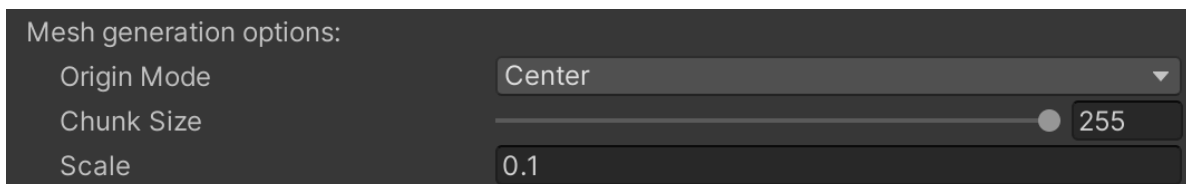


7. **Material Properties Embedding Mode** allows storing surface properties, including color, metallicity, smoothness, and emission, in a compact texture. This differs from the Vertex Mode, where these properties are stored directly within the mesh vertices.

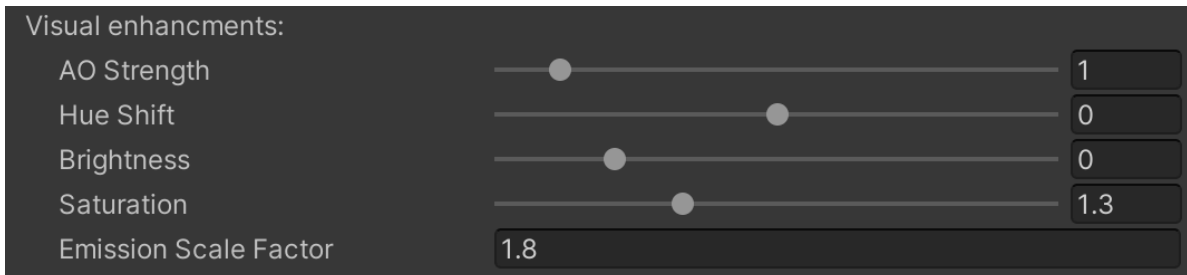


8. **Texture Optimization Mode** defines how the texture is optimized:

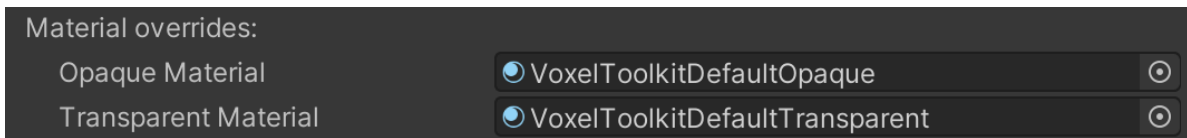
- **None:** No pattern matching is performed.
- **Rotation:** Similar patterns are searched in four rotations (0°, 90°, 180°, 270°).
- **Mirroring:** Pattern matching is performed across four mirror transformations (none, horizontally, vertically, both axes).
- **All:** Pattern matching is performed across all the described transformations.



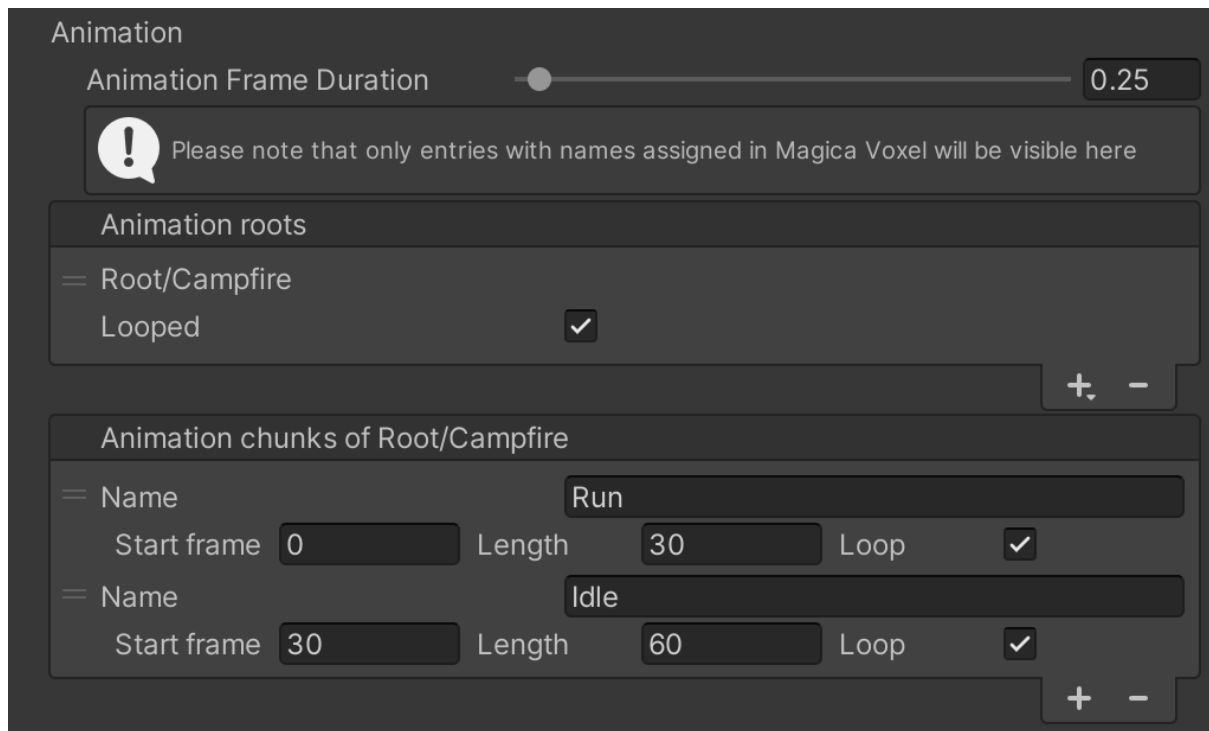
9. **Origin Mode** determines the selection method for the model's origin. If set to 'corner,' the origin will be at the minimum position of the model's bounding box. If set to 'center,' the origin will be at the center of the bounding box.
10. **Chunk Size** defines the size of a voxel cube segment when generating the mesh. For example, a value of 8 splits your mesh into chunks of size 8x8x8 voxels, which are generated separately. The lower the value, the more parallel processing you get and less memory is required during generation, but this might result in less optimal mesh quality. A value between 32 and 128 generally covers 99% of use cases.
11. **Scale** adjusts the size of the imported object, ensuring it integrates seamlessly into your Unity scene with the desired dimensions.



12. **AO Strength.** Determines the intensity of ambient occlusion applied to the model during mesh generation. A value of 0 disables the effect entirely and has no impact on the final result. Enabling ambient occlusion may increase the poly count slightly, as additional geometry can be generated to represent the shading effect.
13. **Hue Shift, Brightness** and **Saturation** refer to the color transformation applied to the voxel colors.
14. **Emission Scale Factor** adjusts the emission parameter due to differences in Unity's rendering system compared to MagicaVoxel, which may result in different visual appearances. To achieve the desired look, we utilize a scale factor to appropriately adjust the emission effect.

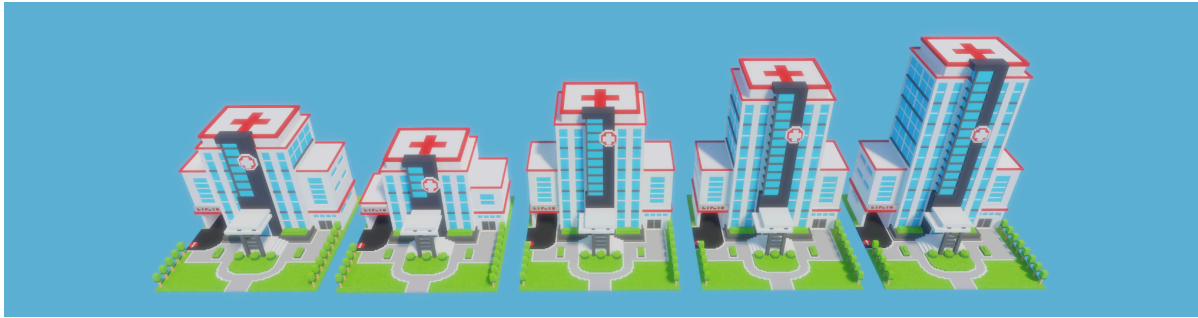


15. **Opaque Material** and **Transparent Material** are the materials that will be used for the generated meshes.

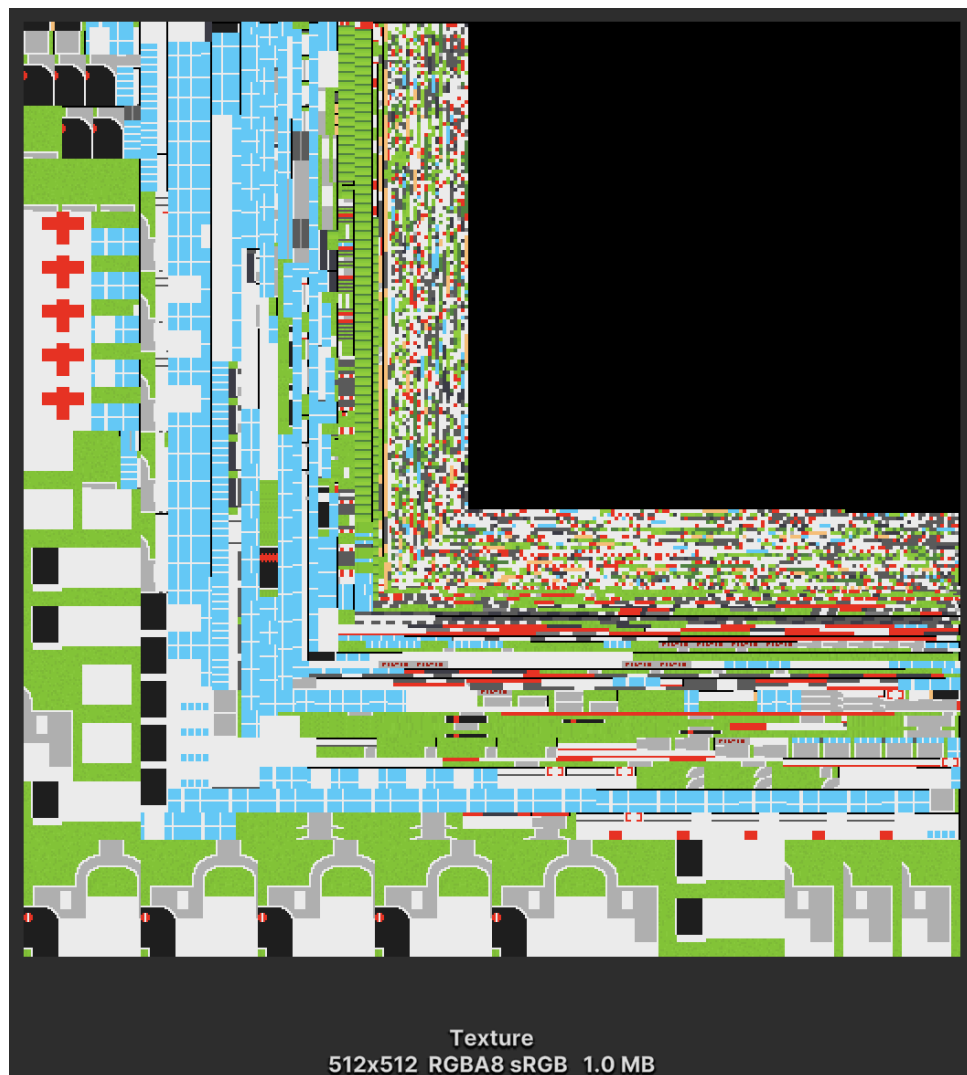


16. **Animation Frame Duration** defines the duration of a single animation frame, in seconds.
17. **Animation roots** specify the objects for which animators will be created after import. Please note that the full hierarchy must be properly named in MagicaVoxel for the root to be detected and selectable.
18. **Animation chunks** represent the list of animations to be created. You can specify whether each animation should loop, assign a name, and define its position within the full MagicaVoxel animation timeline. This is useful for having multiple animations in a single file and for working around MagicaVoxel's limitation of supporting only one animation per file.

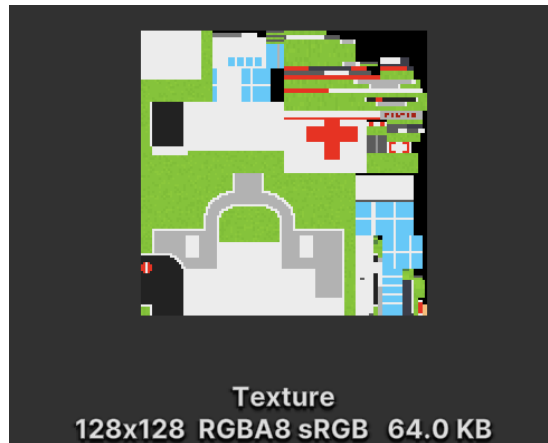
Examples of Texture Compression



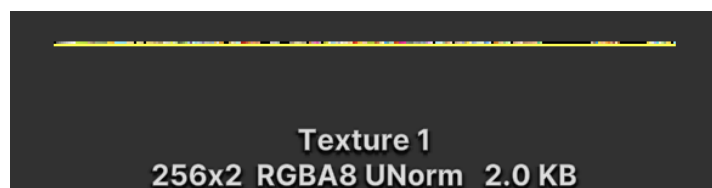
1. **Material Properties Embedding Mode:** Vertex
2. **Texture Optimization Mode:** None



1. **Material Properties Embedding Mode:** Vertex
2. **Texture Optimization Mode:** All



-
1. **Material Properties Embedding Mode:** Texture
 2. **Texture Optimization Mode:** All



Contact Information

1. [Asset Store Page](#)
2. Support Email: pirate.parrot.software@gmail.com
3. [Discord](#) Community

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Links to tutorials and promotional materials:

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For any questions, feature suggestions, or feedback, feel free to contact us through email or join our Discord community! You can also stay updated by following us on social media.

★ Support Us with a Review! ★

If you found this documentation helpful, please take a moment to leave a review! Your feedback is invaluable in helping us improve [Voxel Toolkit: MagicaVoxel Importer & Optimizer](#) and add new features.

Every review makes a difference — thank you for your support!